On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ and showing it is sharp. The lower bound follows from a counting/pigeonhole argument on smooth numbers. The upper bound uses Hildebrand's 1986 theorem on smooth numbers in short intervals.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n, a *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Equivalently, none of $m+1, \ldots, m+k$ is k-smooth. Define $k(n)$ to be the maximum k for which a k-run below n exists.

Remark 3. A k-run below n is exactly a gap of length $\geq k$ in the set of k-smooth integers inside $[1, n+k]$: an interval $[m+1, m+k]$ with $m \leq n$ that contains no k-smooth integer.

2 Smooth Number Background

Theorem 4 (Dickman [1], de Bruijn [2])**.** *Let $\rho\colon [0,\infty) \to (0,1]$ be the Dickman function, defined by $\rho(u) = 1$ for $u \in [0,1]$ and $u\rho'(u) = -\rho(u-1)$ for $u > 1$. For $u > 0$ fixed and $y = x^{1/u}$,*

$$\Psi(x,y) = x\,\rho(u)\,(1+o(1)) \qquad \text{as } x \to \infty.$$

The same holds uniformly for $1 \le u \le (\log x)^{3/5-\varepsilon}$ (Hildebrand–Tenenbaum [4]).

Lemma 5 (Dickman bound)**.** *For all $u \ge 1$, $\rho(u) \le u^{-u}$.*

Proof. One checks inductively that $\rho(u) \le 1/(u\,\Gamma(u)) \le u^{-u}$; see [5, Ch. III.5]. □

The short-interval smooth number count we need is the following theorem of Hildebrand.

Theorem 6 (Hildebrand [3], Theorem 1)**.** *Fix $\varepsilon > 0$. Uniformly for $x \ge 3$ and*

$$y \;\ge\; \exp\!\big((\log\log x)^{5/3+\varepsilon}\big), \tag{1}$$

setting $u = \log x/\log y$,

$$\Psi(x+y,\,y) - \Psi(x,\,y) \;=\; y\,\rho(u)\,(1+o(1))$$

as $x \to \infty$ (with $o(1) \to 0$ uniformly in y satisfying (1)).

Remark 7. Theorem 6 is proved via a Perron integral

$$\Psi(x+y,y) - \Psi(x,y) = \frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT} F_y(s)\,\frac{(x+y)^s - x^s}{s}\,ds + O\Big(\frac{x}{T}\Big),$$

where $F_y(s) = \prod_{p\le y}(1-p^{-s})^{-1}$ and the saddle point σ_0 satisfies $-F_y'/F_y(\sigma_0) = \log x$. The Mertens estimate $\sum_{p\le y} p^{-\sigma}\log p \sim \log(1/(\sigma-1))$ for $\sigma \to 1^+$ and $\sigma \sim 1 - 1/\log y$ pins σ_0. The integral over the vertical segment $(x+y)^s - x^s \approx ys\,x^{s-1}$, and the saddle-point evaluation of $F_y(\sigma_0)$ reproduces the Dickman factor $\rho(u)$. The lower bound condition (1) ensures the error terms in the saddle-point analysis are $o(y\rho(u))$. The full technical details are in [3].

3 The Lower Bound

Theorem 8 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \geq \exp\big((\log n)^{1/2-\varepsilon}\big).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$ and $y = k_0$.

Step 1: The smooth numbers are sparse. Write $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon}(1+o(1))$. By Lemma 5,

$$u_0 \log u_0 = (1/2+\varepsilon)(\log n)^{1/2+\varepsilon} \log\log n\,(1+o(1)) \;\geq\; (\log n)^{1/2+\varepsilon/2}$$

for large n, so $\rho(u_0) \leq e^{-(\log n)^{1/2+\varepsilon/2}}$. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$, we have $e^{-(\log n)^{1/2+\varepsilon/2}} \ll 1/k_0$. By Theorem 4,

$$\Psi(n, k_0) \;=\; n\,\rho(u_0)\,(1+o(1)) \;\leq\; \frac{n}{k_0}$$

for all sufficiently large n.

Step 2: Pigeonhole gives an empty block. Partition $\{1, 2, \ldots, \lfloor n/k_0 \rfloor \cdot k_0\}$ into $B = \lfloor n/k_0 \rfloor$ consecutive blocks $I_i = [(i-1)k_0 + 1,\ ik_0]$ of length k_0 each. There are $B \approx n/k_0$ blocks and at most $\Psi(n, k_0) \leq n/k_0 \approx B$ smooth integers spread across them. In fact we showed $\Psi(n, k_0) \ll n/k_0$, so strictly fewer smooth integers than blocks. By pigeonhole, at least one block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. If $I_i = [(i-1)k_0+1,\ ik_0]$ contains no k_0-smooth integer, then $P^+(j) > k_0$ for every $j \in I_i$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \ldots, m+k_0$ all have largest prime factor $> k_0$. This is a k_0-run below n, so $k(n) \geq k_0$. □

Remark 9. The CRT approach—choose distinct primes $q_1, \ldots, q_k > k$ and solve $q_j \mid m + j$ by CRT—gives modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $Q \gg n$ and $m < Q$ need not satisfy $m \leq n$. The CRT method yields only $k(n) \geq (1-o(1)) \log n / \log\log n$ (from $k \log k \leq \log n$), which is far weaker.

4 The Upper Bound

Theorem 10 (Upper bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \le \exp\big((\log n)^{1/2+\varepsilon}\big).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \le n$ contains a k_1-smooth integer, so no k_1-run below n exists.

Step 1: Verification that Theorem 6 applies. We apply Theorem 6 with $x = m \in [n^{1/2}, n]$ and $y = k_1$. The hypothesis (1) requires

$$\log k_1 \;\ge\; (\log\log m)^{5/3+\varepsilon'} \quad \text{for some } \varepsilon' > 0.$$

We have $\log k_1 = (\log n)^{1/2+\varepsilon}$ and $\log\log m \le \log\log n$, so it suffices to show

$$(\log n)^{1/2+\varepsilon} \;\ge\; (\log\log n)^{5/3+\varepsilon'}.$$

For any fixed $\varepsilon' > 0$ and large n, any positive power of $\log n$ grows faster than any power of $\log\log n$, so $(\log n)^{1/2+\varepsilon} \gg (\log\log n)^A$ for every fixed A. In particular hypothesis (1) holds (with room to spare) for all sufficiently large n and all $m \in [n^{1/2}, n]$. Hence Theorem 6 applies with $x = m$, $y = k_1$.

Step 2: The expected smooth count in $[m+1, m+k_1]$ is large. For $m \in [n^{1/2}, n]$, set $u_1 = \log m / \log k_1 \in [(\log n)^{1/2-\varepsilon}/2,\ (\log n)^{1/2-\varepsilon}]$. Theorem 6 gives

$$\Psi(m+k_1,\, k_1) - \Psi(m,\, k_1) = k_1\, \rho(u_1)\, (1 + o(1)).$$

The product $k_1\rho(u_1)$ satisfies

$$\log\big(k_1\, \rho(u_1)\big) = (\log n)^{1/2+\varepsilon} - u_1 \log u_1\, (1 + o(1)).$$

Since $u_1 \le (\log n)^{1/2-\varepsilon}$,

$$u_1 \log u_1 \le (\log n)^{1/2-\varepsilon} \cdot (1/2 - \varepsilon) \log\log n = o\big((\log n)^{1/2+\varepsilon}\big).$$

Hence $\log(k_1\rho(u_1)) = (\log n)^{1/2+\varepsilon}(1 - o(1)) \to +\infty$, so $k_1\rho(u_1) \to \infty$ and the interval $[m+1, m+k_1]$ contains $k_1\rho(u_1)(1 + o(1)) \ge 1$ smooth integers for large n.

Step 3: Small m. For $m < n^{1/2} \ll k_1$, every integer $j \le k_1$ is k_1-smooth (its prime factors are all $\le j \le k_1$). In particular $[m+1, m+k_1] \supseteq \{1, \ldots, k_1\}$

for $m = 0$, and for general $m < k_1$ the interval contains integers $\le k_1$ which are k_1-smooth.

Conclusion. Every interval $[m+1, m+k_1]$ with $m \le n$ contains a k_1-smooth integer. Hence no k_1-run exists below n, and $k(n) < k_1 = e^{(\log n)^{1/2+\varepsilon}(1+o(1))}$. □

5 Main Result and Sharpness

Theorem 11 (Main Theorem)**.** *As $n \to \infty$,*

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

The proposed bound $\log k(n) \le (\log n)^{1/2+o(1)}$ is ***true and sharp****.*

Proof. Theorems 8 and 10. □

Proposition 12 (Precise asymptotics)**.** $\log k(n) \sim \left(\frac{1}{2}\log n \cdot \log\log n\right)^{1/2}$ *as $n \to \infty$.*

Sketch. The threshold value of k is determined by the self-consistency equation $\rho(u) \asymp 1/k$ with $u = \log n/\log k$. Using $\rho(u) = e^{-u\log u(1+o(1))}$, writing $L = \log k$ and $\ell = \log n$, this becomes $u \log u \sim L$, i.e. $(\ell/L)\log(\ell/L) \sim L$, giving $L^2 \sim \ell\log(\ell/L)$. Setting $L = c\sqrt{\ell\log\ell}$ and expanding: $c^2\ell\log\ell \sim \ell \cdot \frac{1}{2}\log\ell$, so $c = 1/\sqrt{2}$. Hence $\log k(n) \sim \left(\frac{1}{2}\log n \cdot \log\log n\right)^{1/2}$. □

Remark 13 (Primitive sets)**.** Each k-run $\{m+1, \ldots, m+k\}$ is a primitive set: if $a \mid b$ with $a < b$ both in the run, then $b \ge 2a > m+k$, a contradiction. The condition $P^+(a) > k$ gives each element a prime certificate $> k$, making these runs a special class of primitive sets.

References

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